Venkata Lakshmi Parimala Pasupuleti

Senior Python Software Engineer · Python · FastAPI · ML Analytics · Income Verification · Java · AWS · PostgreSQL · 9yr Production

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Senior Python software engineer with 9+ years of production experience building analytic software, ML pipelines, and income verification systems - the exact combination Freddie Mac's Credit Innovation & Analytics team requires. At Entrata, I owned the income verification pipeline for affordable housing for 7 years: HUD federal income limit data imported and validated against 180,000 domain records, driving rent and eligibility calculations for 100K+ housing units - the same automated borrower capacity assessment problem CIA solves at mortgage scale. I implemented business rules in complex compliance software where incorrect income calculations mean incorrect eligibility decisions and regulatory violations - zero tolerance for errors, exactly the stakes of mortgage credit automation. I build and test analytic software end-to-end: Python (9yr production), FastAPI/Flask REST APIs (4+ years), Pandas/NumPy data pipelines, scikit-learn and PyTorch ML models (R^2=0.977 across 190+ country datasets), Java, PostgreSQL/MySQL/Oracle, Docker, AWS (S3/EC2/RDS/Glue), and full CI/CD with automated unit, integration, and regression test suites. I also build ML-powered evaluation systems in production: AgentProbe (5-metric evaluation harness, LLM-as-Judge scoring) and Leasel (8-agent income and compliance analytics pipeline, 93.6% precision across 1,240 test cases). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & APIs: Python (9yr), Java, SQL, PL/SQL, Bash/Shell Scripts, TypeScript, JavaScript, FastAPI, Flask, REST APIs, Linux (9yr production), Git, Make, pip, Conda, Jira, Confluence, Gitlab

Analytics & ML: Pandas, NumPy, scikit-learn, PyTorch, Statistical Modeling, Regression Analysis, Random Forest, Isolation Forest, Machine Learning, ML Model Implementation, Data Analysis, Income Verification, Business Rules Implementation, Automated Testing (pytest, unittest), Full Life Cycle Software Development

Infrastructure & Data: AWS (S3, EC2, RDS, Glue, EKS), Docker, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, GitHub Actions CI/CD, Jenkins, Performance Engineering, Observability, PHI/HIPAA Compliance, LangGraph, LangChain, ChromaDB, Agile/Scrum

PROJECTS

Leasel: Income & Compliance Analytics Pipeline - ML Business Rules

ML Analytics · Income Verification · Business Rules · Production

Python · LangGraph · FastAPI · Pandas · scikit-learn · SQLite · pytest · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production analytic system implementing ML-powered income verification and compliance business rules - the same class of problem Freddie Mac's CIA team automates for mortgage borrower capacity. **Business rules:** multi-stage income verification logic with deterministic guardrails; HITL escalation for borderline cases; zero auto-resolution on regulated decisions. **ML analytics:** 8-agent LangGraph pipeline with conditional routing, confidence scoring, and evidence-backed outputs; 93.6% precision, 87.9% F1 across 1,240 test cases. **Testing:** full pytest regression suite with CI integration blocking quality drift before deployment.

AgentProbe: Analytic Evaluation Platform - Automated Testing & Metrics

Evaluation · Automated Testing · FastAPI · Python · Open Source

Python · FastAPI · LangGraph · ChromaDB · pytest · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production evaluation and automated testing platform built with the same software quality standards analytic systems require. **Testing framework:** 5-metric evaluation engine (correctness, groundedness, hallucination rate, latency, token efficiency); automated unit and regression test suites; CI-integrated gate blocking quality regressions before deployment. **API design:** 15-endpoint FastAPI backend with clean abstractions, multi-provider routing, and secrets management. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable.

AI on FHIR: ML Analytics System - Python, REST APIs & Data Pipelines

ML · REST APIs · Python · Data Analysis · HIPAA

Python · Flask · Java · React.js · TypeScript · Docker · BERT · spaCy · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant analytic system integrating ML with enterprise REST APIs - Python/Flask backend, Java integration layer, React.js + TypeScript frontend, Docker containerization; NLP-powered query translation reducing manual retrieval time ~60%; RBAC with authentication and full data isolation; complete audit trail. Demonstrates the full-stack, full-lifecycle analytic software delivery pattern CIA requires.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Analytic Software Production Delivery

- Designed and delivered a production analytic AI system (Python, LangGraph, FastAPI) end-to-end - requirements, system design, ML business rules implementation, automated testing strategy, and production deployment in weeks as sole technical owner; built HITL guardrails and validation gates for regulated decision paths.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML analytics pipelines (PyTorch, scikit-learn, Random Forest, Isolation Forest, R^2=0.977) on WHO/UK datasets (190+ countries, 100K+ records); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and analytics; translated complex model outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Income verification and analytics software (7 years)** - owned HUD federal income limit pipeline (Python, AWS Glue/S3/RDS): federal income data imported, validated, and transformed into 180,000 domain records driving rent calculations and eligibility decisions for 100K+ housing units; implemented complex business rules in compliance software where incorrect income data triggers regulatory violations; RabbitMQ event-driven async processing with retry logic and idempotency; 99.5% uptime.
- **Full life cycle analytic software development** - Python/FastAPI/Flask REST APIs (4+ years production); Pandas/NumPy data processing pipelines; PostgreSQL/MySQL/Oracle (200+ SQL optimizations, 30% performance gain via execution plan analysis, indexing, partitioning); Docker microservices (50% scalability improvement); Java integration; GitHub Actions CI/CD with automated unit, integration, and regression test suites; 20+ production modules across 120+ Agile sprints.
- **Testing, automation, and quality** - automated testing strategies and tools across all modules; internal debugging tools reducing triage time 40%; bug count maintained below 10 for 6+ consecutive months; Python/Bash/PL-SQL automation eliminating 80% of manual workload; 300+ peer code reviews; mentored 5+ engineers on software quality and production practices.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend analytic modules (Python, Java, PostgreSQL) for production SaaS; income verification data pipeline for 1.6M records across 5 program years; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017